

BS – 101 Engineering Physics

Lecture – 8

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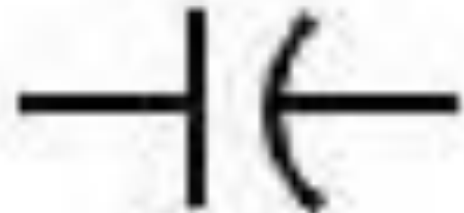


Capacitors

- When you stretch the rubber band of a slingshot or pull back the string of an archer's bow, you are storing mechanical energy as elastic potential energy. A capacitor is a device that stores *electric* potential energy and electric charge. To make a capacitor, just insulate two conductors from each other. To store energy in this device, transfer charge from one conductor to the other so that one has a negative charge and the other has an equal amount of positive charge. Work must be done to move the charges through the resulting potential difference between the conductors, and the work done is stored as electric potential energy.
- Any two conductors separated by an insulator (or a vacuum) form a capacitor.

Capacitors (Diagram)

- In circuit diagrams, a capacitor is represented by either of these symbols:



Capacitance

- The electric field at any point in the region between the conductors is proportional to the magnitude Q of charge on each conductor. It follows that the potential difference V_b between the conductors is also proportional to Q . If we double the magnitude of charge on each conductor, the charge density at each point doubles, the electric field at each point doubles, and the potential difference between conductors doubles; however, the *ratio* of charge to potential difference does not change. This ratio is called the **capacitance** C of the capacitor:

- $$C = \frac{Q}{V_b}$$

Capacitance

- $C = \frac{Q}{V_d}$

- The SI unit of capacitance is called one **farad** (1 F), in honor of the 19th-century English physicist Michael Faraday. One farad is equal to one *coulomb per volt* (1 C/V):

$$1 \text{ F} = 1 \text{ farad} = 1 \text{ C/V} = 1 \text{ coulomb/volt}$$



Capacitance of Parallel Plate Capacitor

- Derivations



Size of a 1 Farad Capacitor

- The parallel plates of a 1.0-F capacitor are 1.0 mm apart. What is their area?



Properties of a Parallel Plate Capacitor

- The plates of a parallel-plate capacitor in vacuum are 5.00 mm apart and 2.00 square meters in area. A 10.0-kV potential difference is applied across the capacitor. Compute:
 - (a) the capacitance;
 - (b) the charge on each plate; and
 - (c) the magnitude of the electric field between the plates.



Capacitor in Series and Parallel Combination (Example)

- Let $C_1 = 6.0 \mu\text{F}$, $C_2 = 3.0 \mu\text{F}$, and $V_0 = 18 \text{ V}$. Find the equivalent capacitance and the charge and potential difference for each capacitor when the capacitors are connected:
 - (a) in series and
 - (b) in parallel.

Capacitor Network (Example)

- Find the equivalent capacitance of the five-capacitor network shown in **Fig.**

